Part Number	41-2287-SB	**Document No.**	WI-0417
Part Name	Steel Mounting Bracket, Stamped	**Revision**	A
Operation	Op 60: Final Assembly and Torque	**Effective Date**	2026-10-05
Work Center	Assembly Cell 4	**Takt / Cycle Time**	3.5 min
Prepared By	Manufacturing Engineering	**Approved By**	Quality Manager

1. Purpose and Scope

This work instruction defines the final assembly of the stamped steel mounting bracket 41-2287-SB, including bushing installation, fastener torque, and final visual inspection prior to pack-out. It applies to all operators certified on Assembly Cell 4.

2. Safety Requirements and PPE

Requirement	Detail
PPE	Safety glasses with side shields, cut-resistant gloves (ANSI A4), steel-toe footwear.
Pinch points	Keep hands clear of the bushing press nest during ram advance. Two-hand anti-tie-down controls must both be used.
Sharp edges	Stamped edges may be sharp. Handle parts by the flange faces only.
Ergonomics	Use the tilt cart for tote changes. Do not lift totes above 35 lb alone.

3. Tools, Equipment, and Materials

Item	Description	Spec / Calibration
Bushing press	Pneumatic press, Station 4-2, with 41-2287 nest	Daily setup verification
DC torque tool	Nutrunner, 10 to 50 Nm range	Cal due date on tool, verify green
Go / No-Go gage	Hole position check fixture G-4128	Gage ID matches route sheet
Torque wrench (audit)	Click-type, 5 to 60 Nm	Calibrated, audit use only

4. Components / Materials Required

Component	Part / Spec No.	Qty
Bracket, stamped and e-coated	41-2287 (Op 50 complete)	1
Rubber isolation bushing	41-2291-RB	2
Flange bolt, M8 x 1.25 x 25, Cl 10.9	F-8825-10	2
Weld nut (verify present from Op 40)	N-8125-W	2

5. Work Steps

Step	Instruction	Key Point / Specification
1	Pull one bracket from the inbound tote. Verify the lot tag on the tote matches the work order on the router.	Lot number matches work order
2	Visually inspect the e-coat surface for scratches, runs, or bare metal. Rotate the part under the cell light.	No bare metal visible

Step	Instruction	Key Point / Specification
3	Verify both weld nuts are present and that threads are clean. Run a thumb test; threads must be free of e-coat buildup.	2 weld nuts, clean threads
4	Load the bracket into the bushing press nest with the flange face down. The part seats flat with no rock.	Part fully seated in nest
5	Place one rubber bushing on each pilot, lip facing up, and confirm orientation against the photo card at the station.	Bushing lip faces UP
6	Cycle the press using both palm buttons. Hold until the ram returns fully.	Press force 6 to 9 kN (auto-monitored)
7	Inspect both bushings for full seating. The bushing shoulder must sit flush against the bracket face with no gap.	Flush, no visible gap
8	Move the bracket to the torque fixture and clamp.	Clamp light green
9	Install both M8 flange bolts finger-tight through the bushing bores into the weld nuts.	Bolts start by hand, no cross-thread
10	Torque both bolts with the DC nutrunner in the sequence marked on the fixture (1 then 2).	24 +/- 2 Nm, tool OK light
11	If the tool faults (red light), back the bolt out fully, inspect threads, and re-run once. A second fault is a nonconformance.	Max 1 retry per joint
12	Check hole position with gage G-4128. The gage must drop fully on both datum pins.	Go condition on both pins
13	Apply the paint pen dot to each torqued bolt head.	White dot, both bolts
14	Place the finished bracket in the outbound dunnage, max 20 per layer, with foam sheet between layers.	20 per layer max

6. Quality Checks

Characteristic	Specification	Method	Frequency
Bolt torque (audit)	24 +/- 2 Nm	Click wrench audit	1 per 50 pcs
Bushing seating	Flush, 0 mm gap	Visual + feeler 0.1 mm	Every part
Hole position	Gage G-4128 Go	Attribute gage	Every part
E-coat coverage	No bare metal	Visual	Every part
First piece	All above	Full check + log	Start of shift

7. Nonconforming Material

If any check fails or a defect is observed, STOP the operation. Tag the part with a red NONCONFORMING tag, record the part number, lot, and defect on the tag, and move the part to the quarantine rack. Notify the line lead and Quality immediately. Do not rework or scrap without an approved disposition.

8. Revision History

Rev	Date	Description	Approved
A	2026-10-05	Initial release	QM

Operator Signature: _____ Date: _____

Supervisor Verification: _____ Date: _____